

A Practical Study of Cryptography in HTTP and HTTPS Communication

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1. Objective

The objective of this practical is to understand the importance of cryptography in web communication by comparing HTTP and HTTPS traffic using Wireshark. The experiment focuses on how sensitive information is transmitted and protected in both protocols.

2. Introduction

HTTP and HTTPS are protocols used for communication between a web browser and a web server. HTTP transfers information without encryption, while HTTPS uses **TLS (Transport Layer Security)** to encrypt the communication.

Wireshark was used in this practical to capture and analyze the packets generated during HTTP and HTTPS communication. By comparing the captured packets, the difference in data protection can be clearly observed.

3. Procedure

1. Wireshark was opened and the **eth0** interface was selected.
2. Packet capture was started before accessing the test websites.
3. A test webpage using **HTTP** was opened and dummy login information was entered.
4. The HTTP packets were captured and examined in **Wireshark**.
5. The same process was repeated using an **HTTPS-secured webpage**.
6. The captured HTTP and HTTPS packets were compared to observe how the application data was transmitted.

4. HTTP Traffic Analysis

The HTTP communication was observed through **port 80**. When the captured HTTP POST request was inspected, the submitted form information could be viewed in readable form.

For example, the captured data could contain information such as:

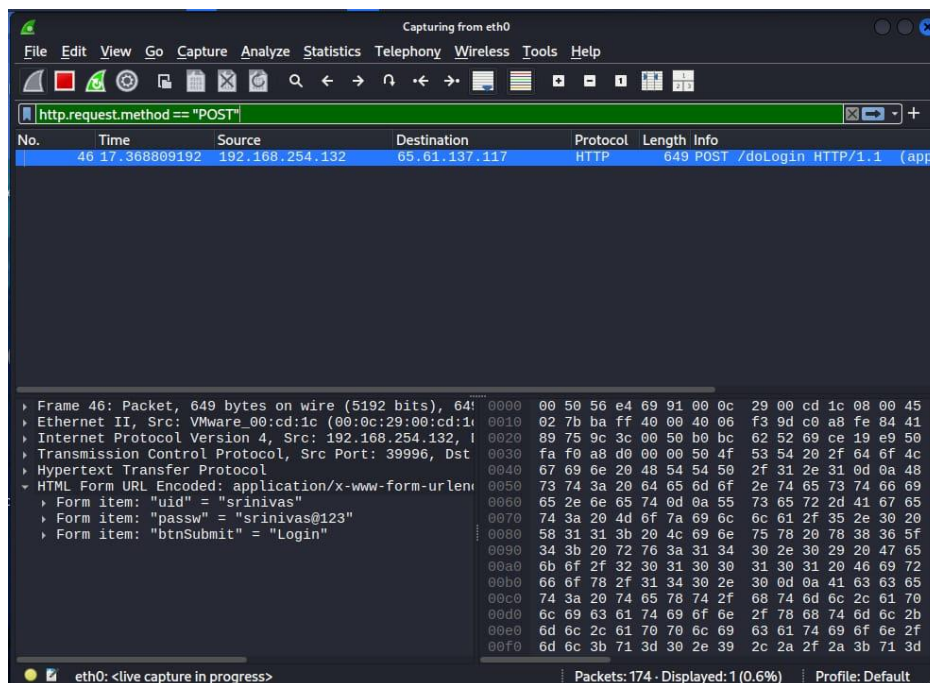
username=Srinivas&password=Srinivas@123

This shows that HTTP does not provide encryption for the application data.

Observation

- Protocol: HTTP
- Port: 80
- Data: Plaintext
- Encryption: Not available
- Security risk: High

If an attacker is able to capture the network traffic, sensitive information such as usernames, passwords, or form data may be exposed.



5. HTTPS Traffic Analysis

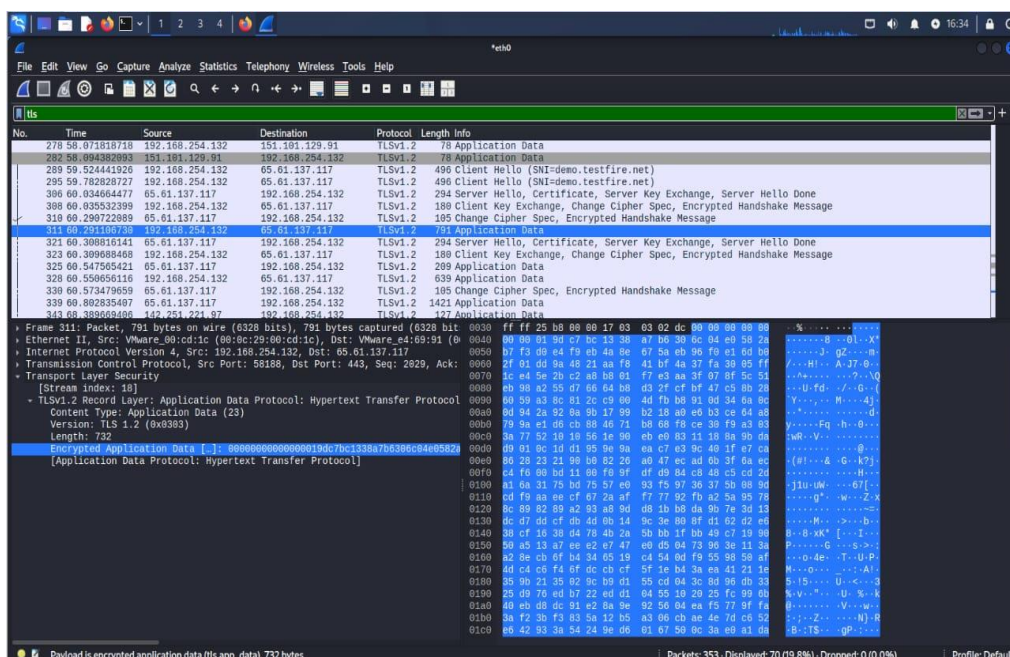
The HTTPS communication was observed through **port 443**. The captured packets contained TLS-protected application data.

Unlike HTTP, the actual submitted information could not be directly viewed in readable form. The packet contained encrypted data instead.

Observation

- Protocol: HTTPS
- Port: 443
- Data: Encrypted
- Encryption: TLS
- Security risk: Much lower

This demonstrates how HTTPS protects sensitive information while it is travelling between the browser and the server.



6. Comparison

Feature	HTTP	HTTPS
Port	80	443
Data format	Plaintext	Encrypted
Encryption	No	TLS
Confidentiality	Not provided	Provided
Integrity	No TLS protection	Protected by TLS
Server authentication	No TLS authentication	Supported
Security	Low	High

7. Role of Cryptography

Cryptography plays an important role in protecting information during network communication.

HTTPS uses TLS to provide:

- **Confidentiality:** Prevents unauthorized users from reading the transmitted data.
- **Integrity:** Helps detect changes made to the data during transmission.
- **Authentication:** Helps verify that the browser is communicating with the intended server.

Therefore, even when HTTPS traffic is captured using Wireshark, the actual application data cannot normally be read directly.

8. Result

The Wireshark experiment clearly showed the difference between HTTP and HTTPS. HTTP traffic allowed application data to be viewed in readable form, while HTTPS traffic protected the application data using TLS encryption.

9. Conclusion

This practical demonstrated why cryptography is important for secure web communication. HTTP does not encrypt sensitive information, making it vulnerable to exposure when traffic is intercepted. HTTPS uses TLS to protect the contents of communication and also provides integrity and server authentication.

Therefore, HTTPS should be preferred over HTTP whenever login credentials, personal information, payment details, or other sensitive data are transmitted.